Newman-Penrose Formalism Calculations

Stephani et al., Equation (23.24)

Metric

$$ds^2 = -\frac{F}{M}\mathrm{d}t \otimes \mathrm{d}t + \frac{F}{N}\mathrm{d}x \otimes \mathrm{d}x + GP\mathrm{d}y \otimes \mathrm{d}y + \frac{G}{P}\mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Metric Functions

$$F = e^{\left(c^2 e^{(2\,au)} + at + \frac{3}{2}\,au\right)}$$

$$u = t + x$$

$$G = e^{(at - ax)}$$

$$M = e^{(-2\,at)} + 1$$

$$N = -e^{(-6\,ax)} + 1$$

$$P = e^{\left(\int \frac{2\,ace^{(ax)}}{\sqrt{N}}\mathrm{d}x\right)}$$

Null Tetrad

Covariant components

$$l = \left(-\frac{\sqrt{2}\sqrt{F}}{2\sqrt{M}}\right) dt + \left(-\frac{\sqrt{2}\sqrt{F}}{2\sqrt{N}}\right) dx$$

$$n = \left(-\frac{\sqrt{2}\sqrt{F}}{2\sqrt{M}}\right) dt + \left(\frac{\sqrt{2}\sqrt{F}}{2\sqrt{N}}\right) dx$$

$$m = \left(\frac{1}{2}\sqrt{2}\sqrt{G}\sqrt{P}\right) dy + \left(\frac{i\sqrt{2}\sqrt{G}}{2\sqrt{P}}\right) dz$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}\sqrt{G}\sqrt{P}\right) dy + \left(-\frac{i\sqrt{2}\sqrt{G}}{2\sqrt{P}}\right) dz$$

Contravariant components

$$l = \left(\frac{\sqrt{2}\sqrt{M}}{2\sqrt{F}}\right) dt + \left(-\frac{\sqrt{2}\sqrt{N}}{2\sqrt{F}}\right) dx$$

$$n = \left(\frac{\sqrt{2}\sqrt{M}}{2\sqrt{F}}\right) dt + \left(\frac{\sqrt{2}\sqrt{N}}{2\sqrt{F}}\right) dx$$

$$m = \left(\frac{\sqrt{2}}{2\sqrt{G}\sqrt{P}}\right) dy + \left(\frac{i\sqrt{2}\sqrt{P}}{2\sqrt{G}}\right) dz$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\sqrt{G}\sqrt{P}}\right) dy + \left(-\frac{i\sqrt{2}\sqrt{P}}{2\sqrt{G}}\right) dz$$

Directional Derivatives

$$D = \frac{\sqrt{2}\sqrt{M}}{2\sqrt{F}}\,\partial_t - \frac{\sqrt{2}\sqrt{N}}{2\sqrt{F}}\,\partial_x$$

$$\Delta = \frac{\sqrt{2}\sqrt{M}}{2\sqrt{F}}\,\partial_t + \frac{\sqrt{2}\sqrt{N}}{2\sqrt{F}}\,\partial_x$$

$$\delta = \frac{\sqrt{2}}{2\sqrt{G}\sqrt{P}}\,\partial_y + \frac{i\sqrt{2}\sqrt{P}}{2\sqrt{G}}\,\partial_z$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\sqrt{G}\sqrt{P}}\,\partial_y - \frac{i\sqrt{2}\sqrt{P}}{2\sqrt{G}}\,\partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\sqrt{N}\frac{\partial}{\partial x}G}{4\sqrt{F}G} - \frac{\sqrt{2}\sqrt{M}\frac{\partial}{\partial t}G}{4\sqrt{F}G}$$

$$\sigma = \frac{\sqrt{2}\sqrt{N}\frac{\partial}{\partial x}P}{4\sqrt{F}P}$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}\sqrt{N}\frac{\partial}{\partial x}G}{4\sqrt{F}G} + \frac{\sqrt{2}\sqrt{M}\frac{\partial}{\partial t}G}{4\sqrt{F}G}$$

$$\lambda = \frac{\sqrt{2}\sqrt{N}\frac{\partial}{\partial x}P}{4\sqrt{F}P}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\sqrt{N}\frac{\partial}{\partial x}F}{8\,F^{\frac{3}{2}}} + \frac{\sqrt{2}\sqrt{M}\frac{\partial}{\partial t}F}{8\,F^{\frac{3}{2}}}$$

$$\gamma = -\frac{\sqrt{2}\sqrt{N}\frac{\partial}{\partial x}F}{8\,F^{\frac{3}{2}}} - \frac{\sqrt{2}\sqrt{M}\frac{\partial}{\partial t}F}{8\,F^{\frac{3}{2}}}$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = {} & \frac{N\frac{\partial}{\partial x}F\frac{\partial}{\partial x}G}{4\,F^2G} - \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial t}F\frac{\partial}{\partial x}G}{4\,F^2G} + \frac{N\frac{\partial}{\partial x}G^2}{8\,FG^2} - \frac{N\frac{\partial^2}{(\partial x)^2}G}{4\,FG} + \frac{\sqrt{M}\sqrt{N}\frac{\partial^2}{\partial x\partial t}G}{2\,FG} - \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial x}F\frac{\partial}{\partial t}G}{4\,F^2G} \\ & + \frac{M\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{4\,F^2G} - \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial x}G\frac{\partial}{\partial t}G}{4\,FG^2} + \frac{M\frac{\partial}{\partial t}G^2}{8\,FG^2} - \frac{M\frac{\partial^2}{(\partial t)^2}G}{4\,FG} - \frac{\frac{\partial}{\partial t}G\frac{\partial}{\partial t}M}{8\,FG} - \frac{\frac{\partial}{\partial x}G\frac{\partial}{\partial x}N}{8\,FG} - \frac{N\frac{\partial}{\partial x}P^2}{8\,FP^2}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = -\frac{N\frac{\partial}{\partial x}G\frac{\partial}{\partial x}P}{4\,FGP} - \frac{\frac{\partial}{\partial x}N\frac{\partial}{\partial x}P}{8\,FP} + \frac{N\frac{\partial}{\partial x}P^2}{4\,FP^2} - \frac{N\frac{\partial^2}{(\partial x)^2}P}{4\,FP}$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{N\frac{\partial}{\partial x}F^2}{8\,F^3} + \frac{N\frac{\partial^2}{(\partial x)^2}F}{8\,F^2} + \frac{M\frac{\partial}{\partial t}F^2}{8\,F^3} - \frac{M\frac{\partial^2}{(\partial t)^2}F}{8\,F^2} - \frac{N\frac{\partial}{\partial x}G^2}{16\,FG^2} + \frac{M\frac{\partial}{\partial t}G^2}{16\,FG^2} - \frac{\frac{\partial}{\partial t}F\frac{\partial}{\partial t}M}{16\,F^2} + \frac{\frac{\partial}{\partial x}F\frac{\partial}{\partial x}N}{16\,F^2} + \frac{N\frac{\partial}{\partial x}P^2}{16\,FP^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = -\frac{N\frac{\partial}{\partial x}G\frac{\partial}{\partial x}P}{4\,FGP} - \frac{\frac{\partial}{\partial x}N\frac{\partial}{\partial x}P}{8\,FP} + \frac{N\frac{\partial}{\partial x}P^2}{4\,FP^2} - \frac{N\frac{\partial^2}{(\partial x)^2}P}{4\,FP}$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = {} & \frac{N\frac{\partial}{\partial x}F\frac{\partial}{\partial x}G}{4\,F^2G} + \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial t}F\frac{\partial}{\partial x}G}{4\,F^2G} + \frac{N\frac{\partial}{\partial x}G^2}{8\,FG^2} - \frac{N\frac{\partial^2}{(\partial x)^2}G}{4\,FG} - \frac{\sqrt{M}\sqrt{N}\frac{\partial^2}{\partial x\partial t}G}{2\,FG} + \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial x}F\frac{\partial}{\partial t}G}{4\,F^2G} \\ & + \frac{M\frac{\partial}{\partial t}F\frac{\partial}{\partial t}G}{4\,F^2G} + \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial x}G\frac{\partial}{\partial t}G}{4\,FG^2} + \frac{M\frac{\partial}{\partial t}G^2}{8\,FG^2} - \frac{M\frac{\partial^2}{(\partial t)^2}G}{4\,FG} - \frac{\frac{\partial}{\partial t}G\frac{\partial}{\partial t}M}{8\,FG} - \frac{\frac{\partial}{\partial x}G\frac{\partial}{\partial x}N}{8\,FG} - \frac{N\frac{\partial}{\partial x}P^2}{8\,FP^2}\end{aligned}$$

$$\Lambda = \frac{N\frac{\partial}{\partial x}F^2}{24\,F^3} - \frac{N\frac{\partial^2}{(\partial x)^2}F}{24\,F^2} - \frac{M\frac{\partial}{\partial t}F^2}{24\,F^3} + \frac{M\frac{\partial^2}{(\partial t)^2}F}{24\,F^2} + \frac{N\frac{\partial}{\partial x}G^2}{48\,FG^2} - \frac{N\frac{\partial^2}{(\partial x)^2}G}{12\,FG} - \frac{M\frac{\partial}{\partial t}G^2}{48\,FG^2} + \frac{M\frac{\partial^2}{(\partial t)^2}G}{12\,FG} + \frac{\frac{\partial}{\partial t}F\frac{\partial}{\partial t}M}{48\,F^2} + \frac{\frac{\partial}{\partial t}G\frac{\partial}{\partial t}M}{24\,FG} - \frac{\frac{\partial}{\partial x}F\frac{\partial}{\partial x}N}{48\,F^2} - \frac{\frac{\partial}{\partial x}G\frac{\partial}{\partial x}N}{24\,FG} - \frac{N\frac{\partial}{\partial x}P^2}{48\,FP^2}$$

Weyl Tensor Components

$$\Psi_0 = \frac{N\frac{\partial}{\partial x}F\frac{\partial}{\partial x}P}{4\,F^2P} - \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial t}F\frac{\partial}{\partial x}P}{4\,F^2P} - \frac{N\frac{\partial}{\partial x}G\frac{\partial}{\partial x}P}{4\,FGP} + \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial t}G\frac{\partial}{\partial x}P}{4\,FGP} - \frac{\frac{\partial}{\partial x}N\frac{\partial}{\partial x}P}{8\,FP} + \frac{N\frac{\partial}{\partial x}P^2}{4\,FP^2} - \frac{N\frac{\partial^2}{(\partial x)^2}P}{4\,FP}$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{N\frac{\partial}{\partial x}F^2}{12\,F^3} + \frac{N\frac{\partial^2}{(\partial x)^2}F}{12\,F^2} + \frac{M\frac{\partial}{\partial t}F^2}{12\,F^3} - \frac{M\frac{\partial^2}{(\partial t)^2}F}{12\,F^2} + \frac{N\frac{\partial}{\partial x}G^2}{12\,FG^2} - \frac{N\frac{\partial^2}{(\partial x)^2}G}{12\,FG} - \frac{M\frac{\partial}{\partial t}G^2}{12\,FG^2} + \frac{M\frac{\partial^2}{(\partial t)^2}G}{12\,FG} - \frac{\frac{\partial}{\partial t}F\frac{\partial}{\partial t}M}{24\,F^2} + \frac{\frac{\partial}{\partial t}G\frac{\partial}{\partial t}M}{24\,FG} + \frac{\frac{\partial}{\partial x}F\frac{\partial}{\partial x}N}{24\,F^2} - \frac{\frac{\partial}{\partial x}G\frac{\partial}{\partial x}N}{24\,FG} - \frac{N\frac{\partial}{\partial x}P^2}{12\,FP^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = \frac{N\frac{\partial}{\partial x}F\frac{\partial}{\partial x}P}{4\,F^2P} + \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial t}F\frac{\partial}{\partial x}P}{4\,F^2P} - \frac{N\frac{\partial}{\partial x}G\frac{\partial}{\partial x}P}{4\,FGP} - \frac{\sqrt{M}\sqrt{N}\frac{\partial}{\partial t}G\frac{\partial}{\partial x}P}{4\,FGP} - \frac{\frac{\partial}{\partial x}N\frac{\partial}{\partial x}P}{8\,FP} + \frac{N\frac{\partial}{\partial x}P^2}{4\,FP^2} - \frac{N\frac{\partial^2}{(\partial x)^2}P}{4\,FP}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"